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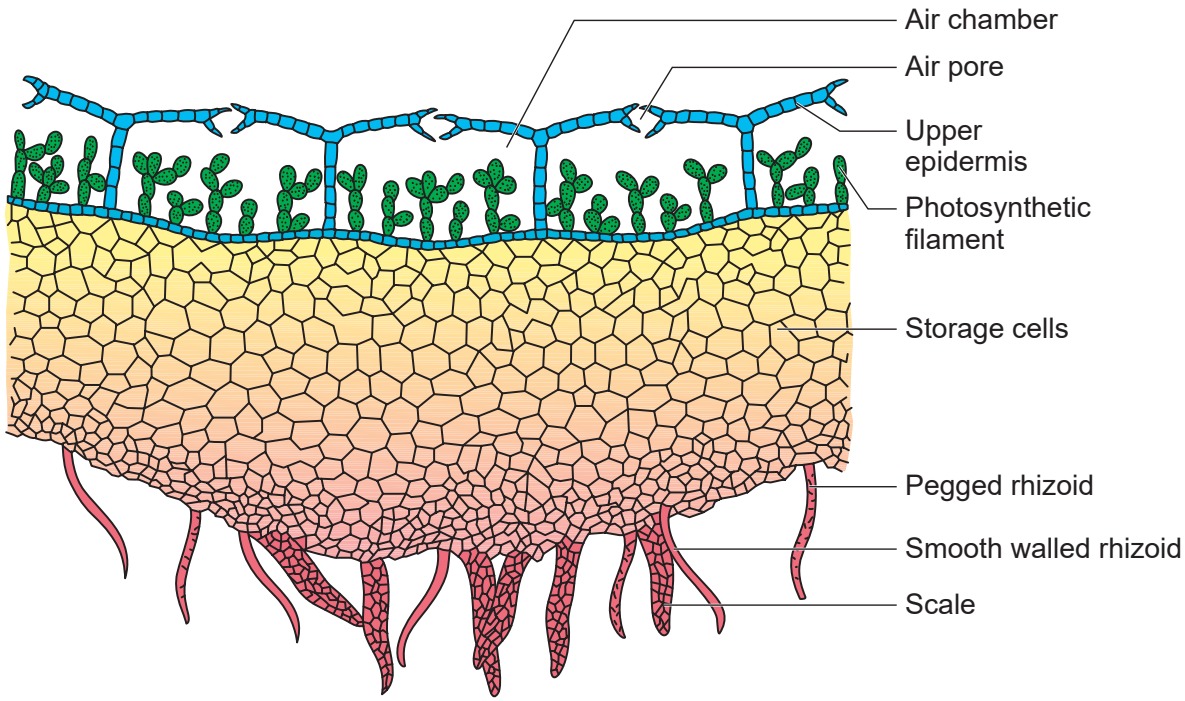
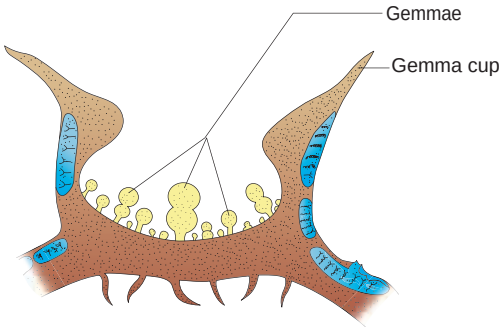
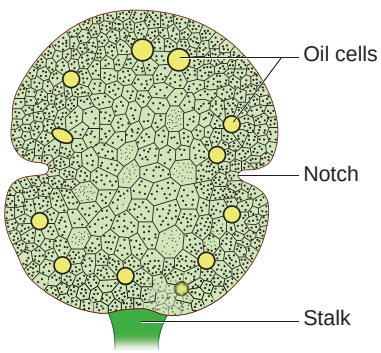


Figure 2.13: T.S. of Thallus

the gemmae present on the male thallus form male plants and those on the female thallus give rise to female plants (Figure 2.14).



a) V.S. of Gemma cup



b) A gemma enlarged

Figure 2.14: Vegetative reproduction in *Marchantia*

1. Sexual reproduction:

In *Marchantia*, sex organs are borne on special stalked receptacles called the gametophores. Those bearing antheridia are called antheridiophores and archegonia bearing structures are called archegoniophores (Figure 2.15). *Marchantia* is heterothallic or dioecious. i.e., male and female receptacles are present on different thalli. The sex organs in bryophytes are multicellular. The male sex organ is called antheridium. It produce biflagellate antherozoids. The female sex organ is flask shaped called archegonium and produces a single egg. Water is essential for fertilization. The antherozoids are released into water and are attracted towards archegonium through chemotaxis. Although many antherozoids enter the archegonium, only one fuses with the egg to form zygote. The zygote represent the first cell of the sporophytic generation. Zygote develops in to a multicellular structure called sporophyte. (Figure 2.16).